

The Fractal Art Maker Algorithm

The Fractal Art Maker's algorithm has two major components: a shape-drawing function and the recursive `drawFractal()` function.

The shape-drawing function draws a basic shape. The Fractal Art Maker program comes with the two shape-drawing functions shown previously in [Figure 13-2](#), `drawFilledSquare()` and `drawTriangleOutline()`, but you can also create your own. We pass a shape-drawing function to the `drawFractal()` function as an argument, just as we passed the match functions to the file finder's `walk()` function in Chapter 10.

The `drawFractal()` function also has a parameter indicating changes to the size, position, and angle of the shapes between recursive calls to `drawFractal()`. We'll cover these specific details later in this chapter, but let's look at one example: fractal 7, which draws a wave-like image.

The program produces the Wave fractal by calling the `drawTriangleOutline()` shape-drawing function, which creates a single triangle. The additional arguments to `drawFractal()` tell it to make three recursive calls to `drawFractal()`. [Figure 13-3](#) shows the triangle produced by the original call to `drawFractal()` and the triangles produced by the three recursive calls.